

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

24th May 2023

**Subject CM2A – Financial Engineering and Loss
Reserving (Paper A)**

Time allowed: 3 Hours 15 Minutes (14.45 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions beginning your answer to each question on a separate sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1)

i) Explain how technical and fundamental analysis relate to the efficient market hypothesis. (4)

ii)

a) Define first and second order stochastic dominance. (2)

b) Highlight the difference between first and second order stochastic dominance. (1)

iii) There are 3 assets A, B and C which give returns as per the distributions below:

Return->	-3.00%	-2.00%	0.00%	2.00%	4.00%
Asset	Probability of Return				
Asset A	0.1	0.2	0.4	0.2	0.1
Asset B	0.2	0.3	0.1	0.3	0.1
Asset C	0.1	0.3	0.2	0.3	0.1

Determine which assets exhibit first and second order stochastic dominance over the others and fill in the table below indicating if the dominance exists, and which asset dominates. (6)

	First Order	Second Order
Assets A and B		
Assets A and C		
Assets B and C		

[13]

Q. 2) Toggle Corporation's share is selling for Rs. 80 per share and its dividend next year is expected to be Rs. 2. Nifty 50 index is 14370 at present and it is expected to go up to 15500 after one year. The average dividend yield for the Nifty 50 index is 1.52% and the risk-free rate is 5.14%. If the beta of Toggle Corporation's share price is 1.14, find the expected share price of Toggle Corporation after one year using CAPM. **[6]**

Q. 3) i) Calculate downside semi variance of an investment with cost 0.7 and having a payoff with probability density function f .

$$f(t) = \begin{cases} \frac{0.375}{t^4}, & t \geq 0.5 \end{cases} \quad (5)$$

ii) List the advantages and disadvantages of using semi variance as a risk measure. (3)

[8]

Q. 4) i) List any 4 types of credit event. (2)

ii) A company has just issued a 3-year zero-coupon bond with a nominal value of Rs. 85 crores. The total Asset value of the company now stands at Rs. 110 crores. Investors use the following annual forward rates to estimate the constant risk free continuously compounded rate.

Year 1	5% p.a
Year 2	5.5% p.a
Year 3	6% p.a

Calculate the theoretical value of Equity and Debt using the Merton model, assuming that the annual volatility of the value of the company's assets is 30%. (7)

[9]

Q. 5) i) Find $E[|W(t)|]$ using first principles if $W(t)$ is a standard Brownian motion. (8)

ii) Share price of ABC Ltd and XYZ Ltd are at the same price at the beginning of a trading day. Let $X(t)$ denote the rupee amount by which Share price of ABC Ltd. exceeds Share price of XYZ Ltd when t percent of the trading day has elapsed.

$\{X(t) \ 0 \leq t \leq 1\}$ is modelled as an Arithmetic Brownian motion process with drift 0 and variance parameter $\sigma^2 = 0.3695$. After 75% of the trading has elapsed, ABC Ltd's share price is Rs. 40.25 and XYZ Ltd's share price is Rs. 39.75. Find the probability that ABC Ltd's share price is higher than XYZ Ltd's share price at the end of the day. (8)

[16]

Q. 6) i) In a Black Scholes market, a European call option has a strike price Rs. 12000 and current stock price is Rs. 11000. The expiry is in 1 year, continuously compounded risk-free rate is 2% p.a. The price of the option is Rs. 100.9. Estimate the implied volatility. (5)

ii) An investment bank writes European call options on a stock with an expiry of six months. The price of this stock on 1-Nov-2022 is Rs. 100 and on 2-Nov-2022 is Rs. 75. But this is expected to increase in the next six months to Rs. 150. The option price on 1-Nov-2022 is Rs. 6 and on 2-Nov-2022 is Rs. 4. Explain why the call option price falls using Black Scholes method despite the expectation that the share price will increase in the future. (2)

iii) Assume a long position in the following options written on the same stock. Compare these in the order of how valuable they are to you.

Option	Type	Strike	Time to expiry
A	American	1000	3 years
B	American	1000	2 years
C	American	1200	3 years
D	European	1000	3 years
E	European	1000	2 years

(5)

iv) Fill in the blanks in the table based on whether the value of the option decreases/increases when each factor increases.

Factors affecting option price	American call	American put
Share price		
Exercise price		
Time to expiry		
Volatility of share price		
Risk free rate of interest		

(5)

[17]

Q. 7) Aggregate claims from a portfolio of insurance policies follows a compound Poisson distribution with Poisson process parameter 25. The individual claim amount follows the following distribution.

Claim	50,000	1,00,000	2,00,000
Probability	30%	50%	20%

Premiums are paid annually in advance. The surplus available is 2,40,000. Calculate the minimum premium loading that would ensure that the probability of ruin at the end of the first year is less than 10%. [6]

Q. 8) i) Explain the following terms in the context of the binomial model:

a) Risk neutral probability measure (3)

b) Recombining binary tree (2)

ii) Consider a 3-period binomial model with following parameters $u = 1.2$, $d = 0.9$, $S_0 = 60$, $r = 11\%$ per period, $K = 60$. Calculate the price of a standard European Call with maturity date at the end of the three periods. (8)

[13]

Q. 9) The paid claims triangle for 3 years for a portfolio of insurance policies is given below. Assume the claims are run off after 3 years.

UWY/ DY	0	1	2	3
2019	10908	13802	15757	16450
2020	9918	12751	14305	
2021	10102	13034		
2022	9679			

i) Estimate the future expected claims from this portfolio. (5)

ii) Construct a table of difference between the actual claims and expected claims that you have estimated using the pattern you have chosen. (4)

iii) Comment on the differences. (1)

iv) State the assumptions in the above calculation. (2)

[12]
